

UCS542: UI & UX SPECIALIST

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Course Objective: The main objective of this course is to impart concepts related with web technology which are essential for the development of web applications. The key technology components include web markup languages, client and server side programming.

Syllabus

HTML& CSS: Introduction to HTML, Introduction, HTML Page Structure, Create HTML document, Understand the various elements available in HTML, HTML Use, Attributes in HTML, Need of Attributes, Common Attributes, HTML forms, Apply validations to the form elements, Creating web pages with HTML5, HTML5 introduced features, HTML5 form validate/no validate, HTML5 canvas, embedding audio, and video in a webpage, drag and drop, HTML5 Local Storage, HTML5 web workers and server sent events. What is CSS, how to insert CSS in HTML, How CSS adds value to HTML, Difference between Semantic and HTML mark-up, CSS3, CSS Selectors, Buttons, CSS float and clear, CSS align - horizontal and center, CSS Padding, CSS Links, CSS Lists, CSS Tables.

JavaScript: What is JavaScript, Importance of JavaScript, What can JavaScript Do?, JavaScript with HTML Attributes, JavaScript with CSS, Operators, JavaScript Syntax, JavaScript Data Types, JavaScript Functions, Setting up Environment, Variables, Control flow, if. Else, switch, loops, JavaScript HTML DOM Elements, JavaScript Syntax, Operators, Data Types, JavaScript String Methods, JavaScript Functions, Arrays, Sorting, Joins, Reduce map.

Frontend Architecture: Introduction to Frontend Development, History, MVC, MVP, MVVM& Web Apps, Development of AJAX, Introduction to DOM, Basic DOM Manipulation, Reactive Programming, Refreshing ES6 Specifications and Features ECMA Script, ES6 let and const, The arrow functions, New Literal Syntax, Classes, Inheritance using extends, Default Parameter Values, Spread Operator (...), Iterators and Generators, Features of React, Practical Application, Why need React, How React Works, Leveraging Virtual DOM, Setting up React.

REST API, JSX: Why JSX, Embedding JavaScript Expression in JSX, JSX Attributes, JSX Comments, Styling and Representation as Object, The State of the Component, Changing the State, Props of Component, Using Props, Props Validation in React, Similarities Between State & Props RESTAPI : Intro to API, History of API Development, Development of AJAX, CRUD; GraphQL; HTTP ,HTTP 1.1,HTTP/2, Stream prioritization, Introduction to React Native, Setting up React Native, The Expo Client, Working up on the First Project, Style, Flexbox Layout.

Node.js: Introduction to Node.js, History, Why Node.js, Node.js Architecture, Features, Working of Node.js, Installation & Setting Up Node, setting up React, REPL Environment, REPL Commands, Variable, Components of Node.js, Local Modules, Module Exports:

Export Object, Export Class, Loading Module from Separate Folder, Operating System, File Systems.

Laboratory Work:

Course Learning Objectives (CLO)

After the completion of the course the student will be able:

1. To acquire knowledge and skills for creation of full web framework considering both client and server side programming.
2. To develop a dynamic and interactive WebPages by the use of HTML, XHTML, CSS, Java

script and DHTML.

3. To build web apps with fully functional frontend with database connectivity.
4. To undertake transfer of multimedia content over the Internet between the multi platforms web applications through communication protocols and the learner will also be able to build web API.

Text Books

1. M. Srinivasan, Web Technology: Theory and Practice, Pearson (2012).
2. A.Kumar, Web Technology: Theory and Practice, CRC Press (2018).

Reference Books:

1. Web programming with HTML, XHTML and CSS, 2e, Jon Duckett, Wiley India.
2. Web technologies: A Computer Science Perspective, Jeffrey C. Jackson ,1st ed., Pearson (2007).